

Week 1 cheat sheet: Python basics

Quantitative Methods in Neuroscience, 2026-27. Keep it next to you while you work.

Running a notebook

Run cell the play button on the left of the cell
Run All runs every cell from the top, in order
Restart throws away every variable the kernel holds
Clear All Outputs removes results and figures, keeps your code
A variable exists only once its cell has been run. If a notebook behaves strangely, **Restart** then **Run All**.

Variables and types

```
n_trials = 240                # int
rt = 0.42                    # float
name = "CSHL-WT"            # str
is_correct = True            # bool, a truth value
nothing = None              # NoneType

type(rt)                    # ask what something is
```

Arithmetic

+ - * / the usual, / always gives a float
// integer division, 7 // 2 is 3
% remainder, 7 % 2 is 1
** power, 2 ** 3 is 8

Text and f-strings

```
label = "mouse " + "CSHL_001"
f"accuracy {acc:.2f}"        # two decimals
f"{n} trials"
len(name)                    # how many characters
name.upper(), name.split("-")
```

Lists

```
rts = [0.31, 0.42, 0.28]
rts[0]                      # first, counting starts at 0
rts[-1]                     # last
rts[0:2]                    # from 0 up to but not including 2
rts.append(0.55)
len(rts), sum(rts), sorted(rts)
0.42 in rts                 # True or False
```

Dictionaries

```
mouse = {"id": "CSHL_001", "lab": "CSHL"}
mouse["id"]                 # look up by key
mouse["age_weeks"] = 14     # add a new one
mouse.keys(), mouse.values()
for key, value in mouse.items():
    print(key, value)
```

Tuples

```
point = (0.25, 0.83)
x, y = point                # unpacking
```

A tuple cannot be changed after it is made, which is why it can be a dictionary key and a list cannot.

Conditionals

```
if contrast == 0:
    print("guess")
elif contrast < 0.25:
    print("hard")
else:
    print("easy")
```

== != equal, not equal
< > <= >= comparisons
and or not combine conditions

Loops

```
for rt in rts:
    print(rt)

for i in range(5):            # 0 1 2 3 4
    print(i)

for i, rt in enumerate(rts):  # index and value
    print(i, rt)

for a, b in zip(labs, groups): # two lists at once
    print(a, b)

while trials_left > 0:
    trials_left = trials_left - 1
```

Reading an error

Read the **last** line first: it names the problem.

NameError a name that was never defined, often a typo
TypeError an operation between incompatible types
IndexError a position that does not exist
KeyError a dictionary or column key that is not there
SyntaxError Python cannot even read the line
IndentationError a block is not indented consistently

Getting help

```
help(round)                 # the full documentation
round?                     # a quick summary, in Jupyter
```

And the official docs: docs.python.org, numpy.org, pandas.pydata.org.

Imports

```
import numpy as np
import pandas as pd
from pathlib import Path
```

A **module** is one file of Python, a **library** is a collection of them.

Files

```
with open("report.txt", "w") as f:
    f.write("one line\n")

with open("report.txt") as f:
    text = f.read()
```

A first look at pandas

```
mice = pd.read_csv(path)
mice.head()                # the first five rows
mice.shape                # (rows, columns)
mice.info()               # column names and types
mice["lab"]                # one column
mice.iloc[0]              # by position
mice.loc[0, "lab"]        # by label
```

Week 3 starts again from the beginning.